

Name: _____ Date: _____

Period: _____

Primary Objective

I will be able to...

1. **LO 3.8.A** — Construct representations of $f(\theta) = \tan \theta$ using the unit circle. _____
2. **LO 3.8.B** — Describe key characteristics of the tangent function (period, asymptotes, monotonicity, concavity). _____
3. **LO 3.8.C** — Describe additive and multiplicative transformations of the tangent function. _____

Vocabulary & Key Concepts

tangent function: The function $f(\theta) = \tan \theta$ that gives the slope of the terminal ray of angle θ on the unit circle.

vertical asymptote: A vertical line $\theta = a$ where the function values grow without bound. For $\tan \theta$, these occur where $\cos \theta = 0$.

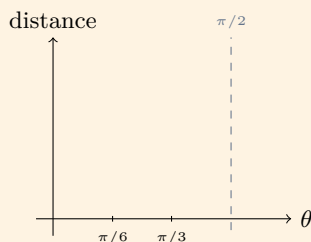
period (tangent): The tangent function has period π because the slope of the terminal ray repeats every π .

Hook

Entry Question: A lighthouse beam rotates at a constant rate. The distance along a straight shoreline to the point illuminated by the beam is a function of the angle θ the beam makes with the perpendicular.

Q1. If the lighthouse is 1 unit from the shore and the angle is $\theta = \pi/4$, how far along the shore does the beam shine? (Hint: think about the tangent ratio in a right triangle.) _____

Q2. What happens as θ approaches $\pi/2$? Sketch your prediction on the axes below.



LO 3.8.A — Constructing $\tan \theta$ from the Unit Circle

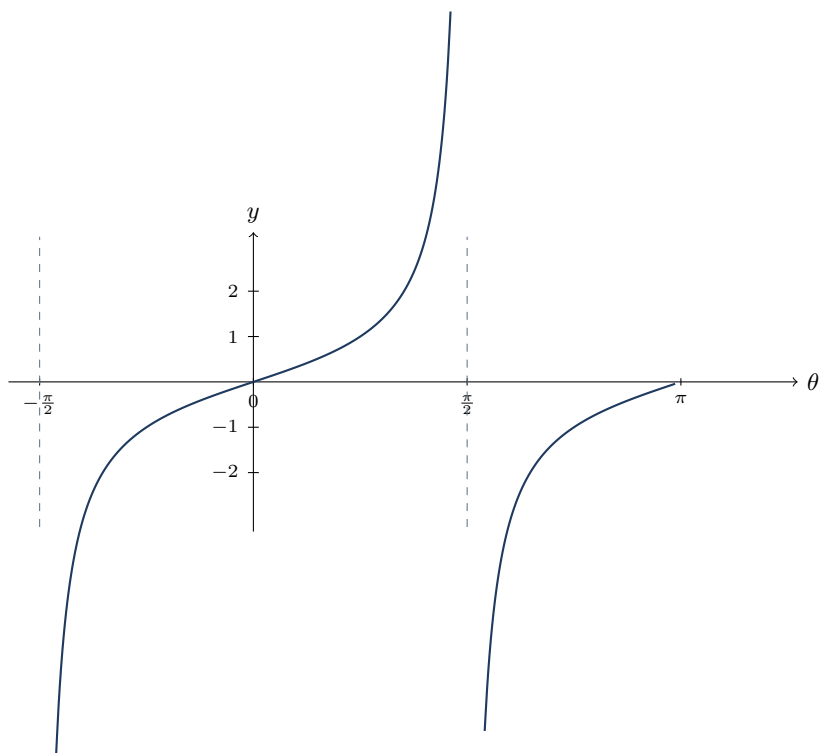
Definition: For any angle θ in standard position, the terminal ray passes through $(\cos \theta, \sin \theta)$. The slope of that ray is:

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \quad \text{provided } \cos \theta \neq 0$$

Complete the table using sin and cos values from your warm-up.

θ	$\sin \theta$	$\cos \theta$	$\tan \theta = \sin \theta / \cos \theta$
0	0	1	_____
$\pi/6$	$1/2$	$\sqrt{3}/2$	_____
$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	_____
$\pi/3$	$\sqrt{3}/2$	$1/2$	_____
$\pi/2$	1	0	_____
$2\pi/3$	$\sqrt{3}/2$	$-1/2$	_____
$3\pi/4$	$\sqrt{2}/2$	$-\sqrt{2}/2$	_____
π	0	-1	_____

Annotate the pre-drawn graph below with the values from your table. Mark with an “ $\times$ ” where vertical asymptotes occur.



LO 3.8.B — Key Characteristics of $y = \tan \theta$

Characteristic	Description / Value
Period	_____
Vertical asymptotes	$\theta = \underline{\hspace{2cm}}$ for integer k
Zeros	$\theta = \underline{\hspace{2cm}}$ for integer k
Monotonicity	The function is _____ on every interval between consecutive asymptotes
Concavity change	Concave _____ for $\theta < 0$; concave _____ for $\theta > 0$ (within $(-\pi/2, \pi/2)$)

Why period = π : After a half-revolution the terminal ray points in the opposite direction, so both rise and run _____, giving the same _____.

LO 3.8.C — Transformations of $y = \tan \theta$

For $g(\theta) = a \cdot \tan(b(\theta + c)) + d$:

Param.	Effect	Example
a	Vertical dilation by _____	$2 \tan \theta$: outputs $\times 2$
b	Period becomes _____	$\tan(2\theta)$: period = $\pi/2$
c	Phase shift of _____	$\tan(\theta - \pi/4)$: shift right $\pi/4$
d	Vertical translation by _____	$\tan \theta + 3$: shift up 3

Finding new asymptotes: Asymptotes occur when the *inside argument* equals $\pm\pi/2 + k\pi$.

Example: Find the asymptotes of $g(\theta) = \tan(2\theta)$.

Set $2\theta = \frac{\pi}{2} + k\pi \Rightarrow \theta = \underline{\hspace{2cm}}$

Guided Practice

Guided Practice. For each function below, state the period and the location of the vertical asymptote closest to $\theta = 0$ (with $\theta > 0$).

Function	Period	Asymptote ($\theta > 0$)
$g(\theta) = \tan(3\theta)$	_____	_____
$g(\theta) = \tan(\frac{1}{2}\theta)$	_____	_____
$g(\theta) = 4 \tan(\theta) + 1$	_____	_____